

Herald



Herald — §11.4.98 Full-Automation Anti-Bluff Audit

Field	Value
Audit date	2026-05-28
Authority	HelixConstitution §11.4.98 (commit 6828ff2) + Herald §108.m (commit bb f03c8)
Scope	All Herald test surfaces (integration tests, shell scripts, mutation gates, e2e invariants)
Release-gate posture	NON-COMPLIANT items have 30 days from this audit before graduating to §11.4.90 Obsolete
Deadline	2026-06-27 (T+30 days)
Methodology	Static analysis only — no test executed, no source modified
Total tests audited	794 (783 Go test/bench/example funcs + 11 shell test/e2e scripts)
COMPLIANT (hermetic)	750 Go funcs + 8 shell scripts = 758
COMPLIANT-with-creds-bootstrap	31 Go funcs + 1 shell script = 32
NON-COMPLIANT-manual-dep	2 Go funcs + 1 shell script + 1 shell script-mode = 4
STRUCTURALLY-BROKEN	1 (carved out of COMPLIANT-with-creds-bootstrap above for clarity)
OBSOLETE-CANDIDATE	0 (no removed code-path tests detected — all NON-COMPLIANT items are rewrite-targets, not deletion-targets)
PLAN-ONLY (out of scope)	0 (no docs/challenges/ dir exists; docs/research/ contains 4 prose-only docs counted out of scope)

Summary

Herald's test surface is overwhelmingly COMPLIANT under §11.4.98: 758 of the 794 audited tests are fully self-driving — they run hermetically with no external dependency or no human action of any kind. A further 32 tests are COMPLIANT-with-creds-bootstrap per the §11.4.98(B) explicit exception: they require one-time credential setup (Telegram bot token + chat id in .env, claude on PATH, Postgres/Redis containers via Podman/Docker) — but once configured, every subsequent run is fully autonomous. This is the §11.4.98(B) permitted carve-out, NOT a manual-dep.

The audit identifies **exactly 4 NON-COMPLIANT items** that require human action during test execution itself — every one of them is a Telegram-inbound-driven test that polls `getUpdates` for a real operator-typed message. These 4 cannot run in unattended CI and are the entire §11.4.98 release-gate debt. They are:

- 1. `TestSubscribe_LiveBotAPI` (`commons_messaging/channels/tgram/subscribe_integration_test.go`) — 60s window for operator hand-send.
- 2. `TestVerticalSlice_TelegramClaudeRoundTrip` (`commons_messaging/vertical_slice_integration_test.go`) — 150s window for operator hand-send, drives the full Telegram → Claude → Telegram round-trip.
- 3. `tests/test_wave6_live_loop.sh` — 60s window for operator hand-send into the configured chat.
- 4. `tests/test_wave6.5_lifecycle.sh --manual (legacy mode)` — 15 narrated scenarios with `read -r` prompts between each one. Note: the DEFAULT mode of the same script delegates to `qa_herald_lifecycle` (a second Telegram bot drives the scenarios) and IS fully self-driving; only the explicit `--manual legacy` path is NON-COMPLIANT.

One additional item is **STRUCTURALLY-BROKEN** (not manual-dep): `TestDispatch_LiveClaudeInvocation` fails when the operator-supplied `HERALD_CLAUDE_SESSION_UUID` collides with the conductor's own active Claude dev session (proven 2026-05-28 in `docs/qa/HRD-LIVE-20260528T082128Z/05_dispatch_live/`). The fix path is different from the manual-dep rewrites: assign a dedicated test-only session UUID, not introduce automation. Carved out separately for transparency.

The next-step plan: **Wave 8 Track B — MTProto user-client harness**. Telegram bots cannot see other bots' messages (proven blocker, `docs/research/telegram-bot-to-bot-constraint.md`), so a 2nd-bot automation driver is impossible. The remaining path is a `gotd/td` user-client that posts messages on behalf of a real Telegram user account; this drives all 4 NON-COMPLIANT items in one fix. Operator-pending: see `docs/requirements/blockers/missing_env_variables.md` for the credentials needed (`TG_API_ID`, `TG_API_HASH`, user phone, session file).

COMPLIANT tests (hermetic — no creds, no human action)

Go tests — fully hermetic (sampled by package)

These are the default-hermetic majority. Every test below runs with `go test ./...` (no `-tags=integration`), no env vars, no human input, no external services. The audit confirms this by absence: each file uses only Go stdlib, `httptest`, in-process loopback, mocks, or the `commons/stresschaos` scaffold. Total: ~750 funcs across the 129 non-integration `_test.go` files.

Test surface	Module / file pattern	Why COMPLIANT
Commons L0	<code>commons/*_test.go</code> , <code>commons/cli/*_test.go</code> , <code>commons/gitops/*_test.go</code> , <code>commons/stresschaos/*_test.go</code>	Pure-Go unit tests. No external dependencies. Uses <code>commons</code> scaffolding for <code>serve_dualstack</code> , <code>h3_test.go</code> , <code>httptest</code> , and <code>process</code> . Gin server runs on local network.

Test surface	Module / file pattern	Why COMPLIANT
Prefix	commons_prefix/prefix_test.go	Algorithm test
Auth	commons_auth/verifier_test.go	JWT verifier against in-test keys.
TLS	commons_tls/cert_test.go	Self-signed cert, parse round-trip
Storage	commons_storage/migrations_test.go, storage_test.go, resource_stress_chaos_test.go	Migration path checks; no DB
Infra (default)	commons_infra/boot_test.go, clients_test.go, task_repository_test.go	Mock-Pool test, _integration_siblings are test default.
Constitution	commons_constitution/audit_test.go, bundle_test.go, cloudevents_test.go, emit_test.go, evaluator_test.go, eventbus_test.go, integration_test.go, ladder/memory_test.go, state/memory_test.go, plus audit_stress_chaos_test.go	In-memory state implementation, commons/scaffold.
Messaging — channels (default)	commons_messaging/channels/channel_test.go, inbox_test.go, registry_test.go, selffilter_test.go, null/null_test.go, slack/*_test.go, tgram/attachments_test.go, tgram/export_test.go, tgram/send_ratelimit_test.go, tgram/send_reply_test.go, tgram/send_security_test.go, tgram/subscribe_test.go, tgram/subscribe_lifecycle_test.go, tgram/webhook_test.go, tgram/stress_chaos_test.go	Hermetic: in-fault-injector chaos suite; no
Messaging — dispatch (default)	commons_messaging/dispatch/claude_code/bootstrap_test.go, claude_code_test.go, dispatch_stress_chaos_test.go, export_test.go	Mock claude via fake binary t.TempDir
pherald	all pherald/cmd/pherald/*_test.go, pherald/internal/{bindings,http,inbound,runner,wizard}/**/*_test.go (43 funcs)	Cobra unit tests, memory dispatch, fake redis/PC, fakes_test
qaherald	all qaherald/cmd/qaherald/*_test.go, qaherald/internal/{herald,lifecycle,messenger,report,scenario,transcript}/**/*_test.go	Unit tests + b harness; the l gated behind checks.
Flavor binaries	{bherald,cherald,iherald,rherald,scherald,sherald}/cmd/*/*_test.go, {...}/internal/**/*_test.go	Cobra unit tests, compliance/b tests, all herm

Shell scripts — fully hermetic (no creds, no human action)

Script	Why COMPLIANT
tests/ test_constitution_inheritance.sh	Parent-walk + grep against checked-in files.
tests/ test_constitution_inheritance_meta.sh	Paired §1.1 mutation gate — snapshots → mutates Constitution.md → asserts gate FAILs → restores. Autonomous.
tests/test_i6_refinement_meta.sh	

Script

tests/test_i8_usability_meta.sh

tests/test_wave2_mutation_meta.sh

tests/test_wave3_mutation_meta.sh

tests/test_wave4_mutation_meta.sh

tests/test_wave4b_mutation_meta.sh

tests/test_wave6_mutation_meta.sh

tests/test_wave6.5_mutation_meta.sh

tests/test_wave7_mutation_meta.sh

tests/
test_stress_chaos_mutation_meta.sh

Why COMPLIANT

Paired §1.1 mutation gate against
.gitmodules. Autonomous.

Paired §1.1 mutation gate against
CLAUDE.md / AGENTS.md /
HERALD_CONSTITUTION.md.
Autonomous.

Paired §1.1 mutation gate against flavor
binaries (sherald/cherald/iherald/
bherald/rherald/scherald) +
e2e_bluff_hunt invariants E19-E33.
Autonomous.

Paired §1.1 mutation gate for Wave 3a
JWT verifier + pherald Runner pipeline
(E35-E47). Autonomous.

Paired §1.1 mutation gate for Wave 4a
HTTP/3 + Brotli + Alt-Svc + TLS-1.3
(E49-E55). Autonomous.

Paired §1.1 mutation gate for Wave 4b
TOON content negotiation (E56-E62).
Autonomous.

Paired §1.1 mutation gate for Wave 6
inbound runtime + CC headless bridge.
Hermetic detectors only (per file
comment: "no live Telegram / claude
binary needed"). Autonomous.

Paired §1.1 mutation gate for Wave 6.5
ticket-lifecycle (T1..T6). Hermetic.
Autonomous.

Paired §1.1 mutation gate for Wave 7
generic messenger framework (registry
+ selffilter + Slack adapter). Hermetic.
Autonomous.

Paired §1.1 mutation gate for the
GAP-3 §11.4.85 stress + chaos suite
(HRD-130, 5 mutations). Hermetic
detectors. Autonomous.

COMPLIANT-with-creds-bootstrap tests (§11.4.98(B) explicit exception)

These require one-time credential bootstrap OUTSIDE test execution. Once configured (.env + ~/.bashrc exports + claude on PATH + container runtime up), every subsequent run is fully autonomous. This is the §11.4.98(B) permitted exception and is **COMPLIANT** per the rule.

Go integration tests (creds-bootstrap, fully autonomous once configured)

Test

TestHealthCheck_LiveBotAPI

File

Test**File**

	commons_messaging/channels/tg/ healthcheck_integration_test.go
TestSend_LiveBotAPI	commons_messaging/channels/tg/ send_integration_test.go
TestSend_PersistsDeliveryEvidence	commons_messaging/channels/tg/ persist_integration_test.go
TestBootstrapSession_LiveClaudeInvocation	commons_messaging/dispatch/ claude_code/ bootstrap_integration_test.go
TestDispatch_PersistsSessionState	commons_messaging/dispatch/ claude_code/ persist_integration_test.go
TestLiveBoot_PostgresOnly	commons_infra/ boot_integration_test.go
TestUp_PopulatesPool	commons_infra/ clients_integration_test.go
TestUp_PopulatesRedis_TTLRoundTrip	commons_infra/ clients_integration_test.go
TestUp_PopulatesQueue_EnqueueDequeueRoundTrip	commons_infra/ clients_integration_test.go
TestRepo*_RoundTrip (×14)	commons_infra/ task_repository_integration_
TestRLS_TenantIsolation_RoundTrip	commons_storage/ storage_integration_test.go
TestPostgresStore_RecordAndGet + 5 siblings	commons_constitution/ postgres_integration_test.go
TestClient_LiveGetMe	qaherald/internal/tgram/ client_test.go
TestTgram_Chaos_GetUpdatesPollerResilience	

Test

File

commons_messaging/channels/t
stress_chaos_test.go

Shell script (creds-bootstrap, fully autonomous once configured)

Script

Bootstrap requirement

**Why
COMPLIANT**

tests/

test_resource_stress_chaos.sh

Podman/Docker + opt-in

HERALD_STRESS_LIVE_OOM=1

The opt-in flag is a host-safety gate (§12.6) — not a manual-dep during execution.

Once set, the script runs container OOM + connection-churn scenarios autonomously. SKIPS-with-reason if creds/flag absent (per §11.4.3).

DEFAULT mode delegates the 15-scenario lifecycle to qaherald lifecycle — the qa-bot posts each scenario's input via Telegram, reads pherald-bot's replies, asserts wire-byte evidence, all without human typing.

tests/

test_wave6.5_lifecycle.sh
(DEFAULT mode)

HERALD_TGRAM_BOT_TOKEN +
HERALD_TGRAM_CHAT_ID +
HERALD_QA_BOT_TOKEN (2nd
Telegram bot) +
HERALD_OPERATOR_IDS +
claude on PATH

The - - manual flag flips to the legacy operator-typing path (see NON-COMPLIANT below). The - - manual flag flips to the

Script	Bootstrap requirement	Why COMPLIANT
		legacy operator-typing path (see NON- COMPLIANT row below).

E-invariants in scripts/e2e_bluff_hunt.sh (88 total, E1-E88)

scripts/e2e_bluff_hunt.sh is the canonical end-to-end smoke. Of its 88 invariants (E1-E88):

- **E1-E12, E19-E33, E35-E48, E49-E62, E81-E88** (~70 invariants): hermetic or creds-bootstrap. COMPLIANT.
- **E13-E18** (6 invariants): creds-bootstrap (live PG / Redis / Telegram outbound / Claude Code). COMPLIANT.
- **E34** (vertical slice): manual-dep — wraps `TestVerticalSlice_TelegramClaudeRoundTrip`. The e2e harness SKIPS E34 unless `HERALD_TGRAM_LIVE_INBOUND=1` is set (signalling an attended session). Classified NON-COMPLIANT via the underlying Go test.
- **E63-E70** (Wave 6 inbound): currently SKIP-with-reason pending T10b live closed-loop evidence; E70 explicitly gated behind `HERALD_W6_LIVE_LOOP=1` because `tests/test_wave6_live_loop.sh` is an ATTENDED test. Classified NON-COMPLIANT via the underlying script.
- **E71-E80** (Wave 6.5 lifecycle): currently SKIP-with-reason pending the full S1..S15 evidence dir. The new automated mode (qaherald lifecycle) will close these once an operator runs it; until then SKIP.

NON-COMPLIANT-manual-dep tests (require human action during execution)

Test	File
TestSubscribe_LiveBotAPI	commons_messaging/channels/tgram. subscribe_integration_test.go (lines 30-58)
TestVerticalSlice_TelegramClaudeRoundTrip	commons_messaging/ vertical_slice_integration_test.go (lines 70-280)

Test

File

tests/test_wave6_live_loop.sh

tests/test_wave6_live_loop.sh (lines 53-58 narrate the precondition; lines 64-92 poll for the message)

tests/test_wave6.5_lifecycle.sh --manual (legacy mode only)

tests/test_wave6.5_lifecycle.sh (lines 156-346 — --manual mode; line 210 read -r _ after each narrate prompt; 15 such prompts S1..S15)

STRUCTURALLY-BROKEN (carved out for clarity — not manual-dep)

Test

File

Failure mode

TestDispatch_LiveClaudeInvocation

commons_messaging/dispatch/
claude_code/
dispatch_integration_test.go
(lines 30-116)

FAILs with exit code 1 on stdout when operation HERALD_CLAUDE equal to the conducting dev-session UUID 2026-05-28 in do LIVE-2026052005_dispatch_ after 180.02s).

OBSOLETE-CANDIDATE (none detected)

No NON-COMPLIANT test exercises a removed or superseded code path. All 4 NON-COMPLIANT items + the 1 STRUCTURALLY-BROKEN item target production code that is live and load-bearing (`tgram.Subscribe` long-poll loop, `claudcode.Dispatch`, `pherald listen` inbound runtime, the full §32.6 ticket-lifecycle pipeline). The rewrite path is **swap the driver, keep the assertions** — not delete-and-replace.

If, after Wave 8 Track B lands, any of the legacy operator-typing prompts in `tests/test_wave6.5_lifecycle.sh --manual` is decided to be a maintenance burden, it should graduate to §11.4.90 Obsolete (→ `Fixed.md`) with Reason superseded-by-design-change and Superseding-item pointing at the qaherald-driven automated mode.

PLAN-ONLY (out of scope — for completeness)

`docs/research/` carries 4 prose-only research documents with NO executable tests:

- `docs/research/constitution-compliance-audit-2026-05-27.md`
- `docs/research/hrd-obsolescence-and-qa-coverage-audit-2026-05-27.md`
- `docs/research/telegram-bot-to-bot-constraint.md` (the structural blocker cited above)
- `docs/research/workable-items-phase2-assessment-2026-05-27.md`

These emit no PASS/FAIL and are correctly out of §11.4.98 scope. `docs/research/protocols/` (58 sub-docs) is also prose-only and out of scope.

No `docs/challenges/` directory exists in this repo.

Release-gate timeline

Date	Milestone
2026-05-28 (today)	Audit published. NON-COMPLIANT items: 4. Release-gate clock starts.
2026-06-05 (T+8d)	Wave 8 Track B planning checkpoint — MTPProto harness HRD-NNN filed; operator credentials request (<code>TG_API_ID</code> , <code>TG_API_HASH</code> , user phone) acknowledged.
2026-06-13 (T+16d)	Wave 8 Track B implementation midpoint — harness skeleton + 1 of 4 NON-COMPLIANT items rewritten end-to-end (canonical target: <code>TestSubscribe_LiveBotAPI</code>).
2026-06-20 (T+23d)	Wave 8 Track B implementation closeout — all 4 NON-COMPLIANT items rewritten + STRUCTURALLY-BROKEN

Date	Milestone
	TestDispatch_LiveClaudeInvocation fixed (dedicated test-only session UUID).
2026-06-27 (T+30d)	§11.4.98 release-gate deadline. All 4 NON-COMPLIANT items MUST either: (a) be rewritten autonomous and PASS in CI, OR (b) be migrated to §11.4.90 Obsolete (→ Fixed.md) with full forensic detail (Since, Reason, Superseding-item, Triple-check evidence per §11.4.6). No third option exists.

Blocker on the timeline: Wave 8 Track B is blocked on operator-supplied Telegram user-account credentials. The blocker is fully documented at docs/requirements/blockers/missing_env_variables.md. Until those credentials land, the §11.4.3 SKIP-with-reason posture holds — the audit honestly reports the items as NON-COMPLIANT-but-honestly-skipped, not as PASS-bluffs.

Cross-references

Anchor	Path / commit
HelixConstitution §11.4.98 (canonical)	../constitution/Constitution.md commit 6828ff2
Herald §108.m (project-binding restatement)	docs/guides/HERALD_CONSTITUTION.md r8 §108.m, commit bbf03c8
Live-evidence README (2026-05-28 run)	docs/qa/HRD-LIVE-20260528T082128Z/README.md
Missing-env blocker	docs/requirements/blockers/missing_env_variables.md
Telegram bot-to-bot wall (Wave 7→8 driver constraint)	docs/research/telegram-bot-to-bot-constraint.md
§107 covenant (Herald)	CLAUDE.md §"End-user-usability covenant" + docs/guides/HERALD_CONSTITUTION.md §107
§107.x docs/qa evidence mandate	CLAUDE.md §107.x + Helix §11.4.83
§107.y working-tree quiescence (mutation-gate safety)	CLAUDE.md §107.y + Helix §11.4.84
§11.4.90 Obsolete status mechanism	Helix §11.4.90, restated in CLAUDE.md
§11.4.4 test-interrupt-on-discovery	Helix §11.4.4 (PASS-bluff discovery halts the cycle)
§11.4.3 explicit SKIP-with-reason	Helix §11.4.3 — the only honest non-PASS posture in absence of dependency

Audit posture statement. Herald is, to within the 4 NON-COMPLIANT items above + the 1 STRUCTURALLY-BROKEN item, §11.4.98-compliant today. The 5 outstanding items are fully scoped to a single Wave (8 Track B) and a single dependency (operator-supplied Telegram user-account credentials). No structural rearchitecture is required; the rewrites swap drivers and keep assertions. The release-gate clock is honest: 30 days, no extension hatch, two paths (rewrite OR §11.4.90 Obsolete). The DEFAULT path of tests/test_wave6.5_lifecycle.sh (automated qaherald-driven) is COMPLIANT today; only its explicit legacy --manual mode counts toward the debt.

— end of audit —